

Bocconi

COMPUTER SCIENCE

code 30424 a.y. 2024-2025

Lesson 12

Introduction to coding



Università
Bocconi
MILANO

Objectives of the lesson

- Understand the basic concepts related to programming, its objectives and its importance
- Getting to know Python
- Installing Python
- Typing the first instructions in Python

...But, why?!



**EVERYBODY SHOULD LEARN
HOW TO PROGRAM A COMPUTER
BECAUSE IT TEACHES YOU HOW
TO THINK.**

- Steve Jobs

(1995)

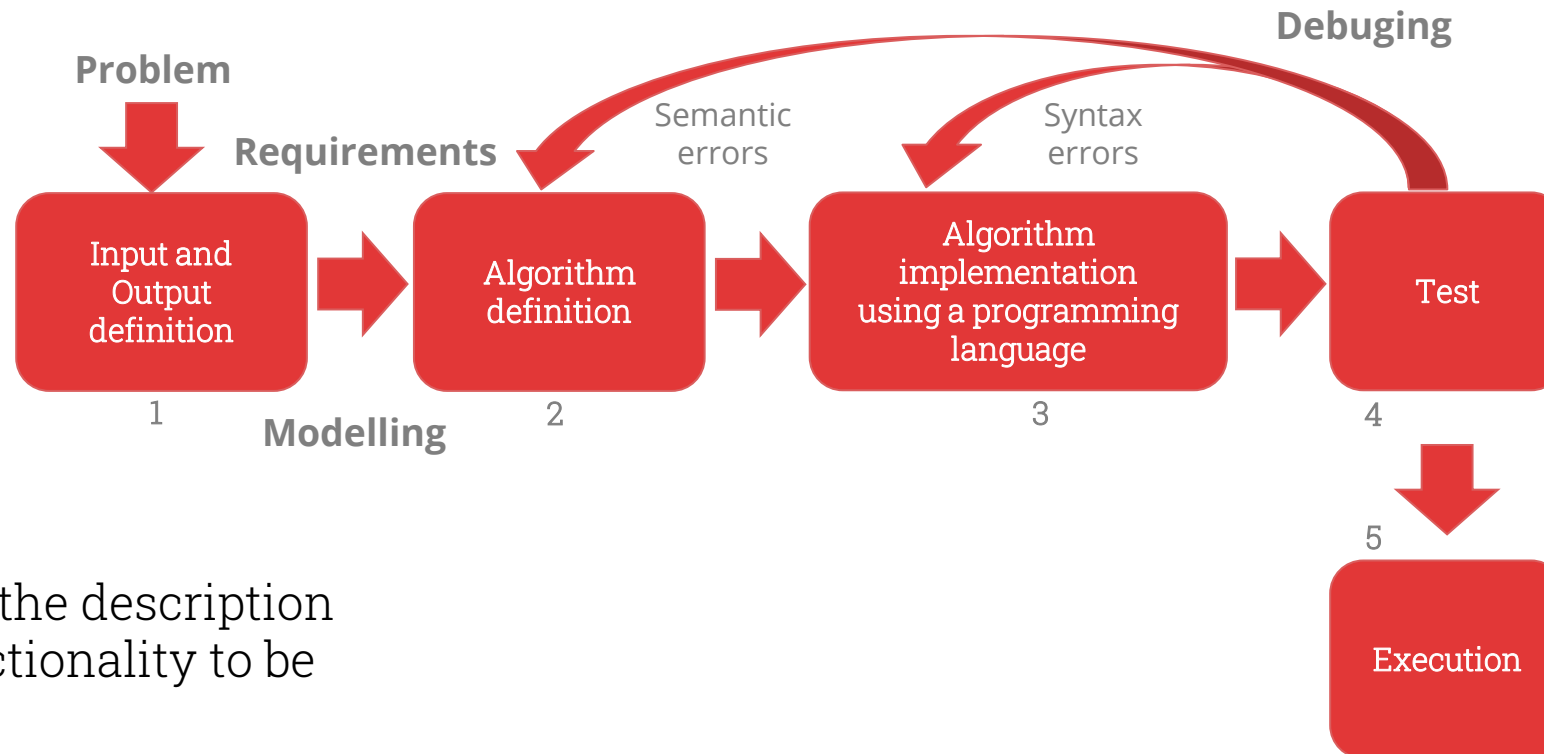
What is programming?

- It is basically teaching a machine how to **perform a certain task or solve a certain problem** (may it be simple or very complex)
- To do it, the computer needs **precise instructions**
- To provide these instructions the programmer must possess a series of **technical skills** (know the logic of programming and the specific language) and **soft skills** (computational thinking, problem solving, creative thinking)

Why programming at Bocconi?

- Successful companies are increasingly interested in people who are able to **understand computer programming** and to apply their knowledge to several environments and disciplines
- Knowledge of computer programming is not only useful to programmers, but also for **everyday tasks** and in **support of business decisions**
- Learning the basics may facilitate the approach to other computer science related subjects and activities

The programming process



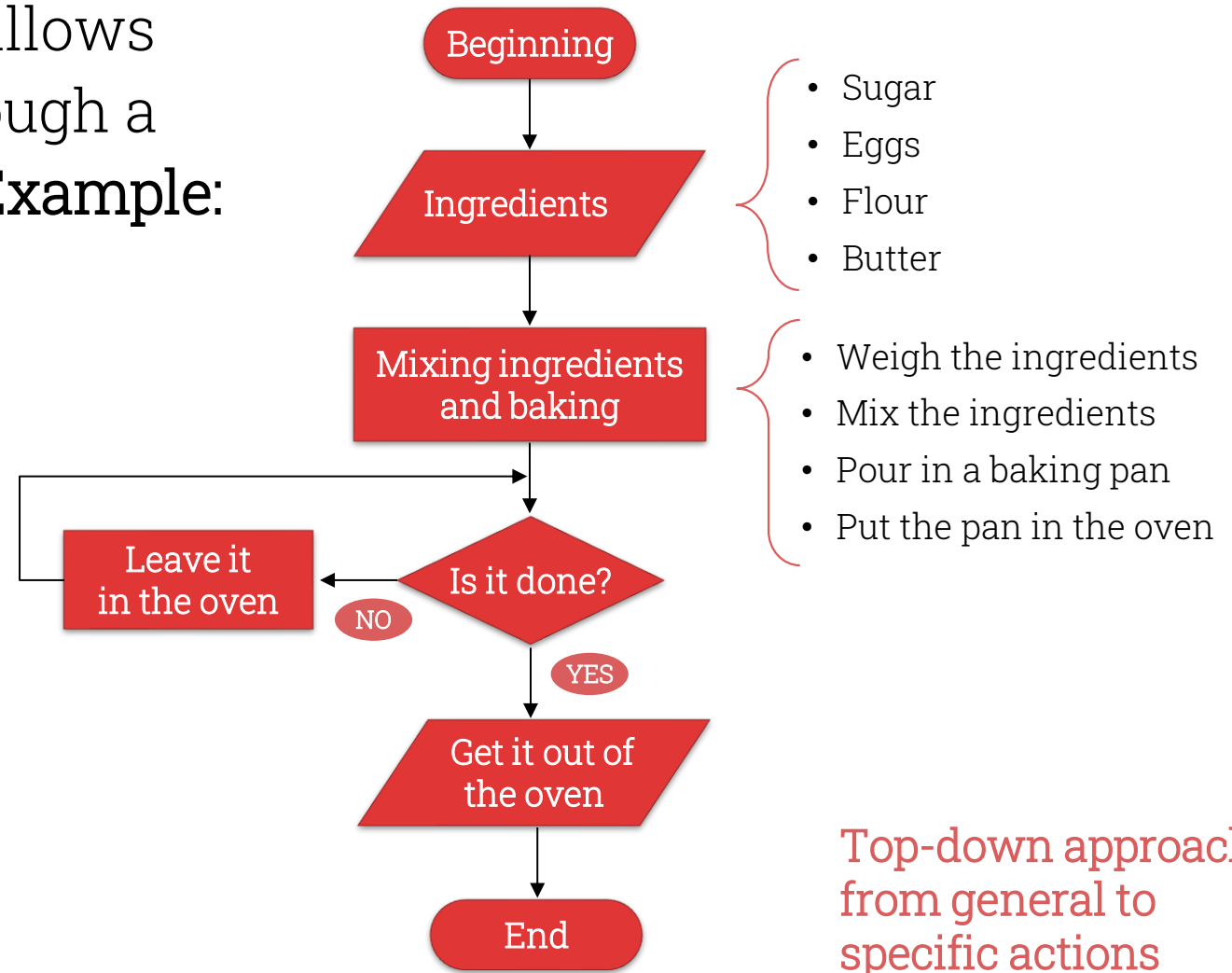
A **Requirement** is the description of a software functionality to be developed

A **Model** is the representation of all the entities and relations among entities for each requirement

The **algorithm** is the process that allows the user to solve a problem through a finite number of simple steps.

Algorithms

An **algorithm** is a process that allows the user to solve a problem through a finite number of simple steps. **Example:** cake recipe



Programming languages

- In Computer science, algorithms are implemented using programming languages, which facilitate interaction between humans and machines, translating instructions and commands in long sequences of bits
- There are many programming languages (both compiled or interpreted), each with its own syntax but basically all providing an interface for giving a computer the instructions needed to solve a problem



Programming languages enable the **communication between man and machine**, like natural languages do between individuals



What is a software program?

It is a sequence of **statements** through which a computer **executes an elaboration** (ex. polynomial equations, mobile apps, autonomous driving cars, ...)

The computer receives **inputs** (from a keyboard, a file, ...) and provides **outputs** (to screen, file, network, ...) by means of **simple operations**:

- mathematical (add, subtract, multiply, ...)
- conditional (if-then-else, ...)
- loops (while, for, ...)

Few LOCs  Billions of LOCs
(LOC = Lines Of Code)

You can think of programming as the process of breaking a large complex task into smaller and smaller subtasks until the subtasks are simple enough to be performed with one of these **basic instructions**

Examples

Software for self-driving cars



100M LOCs



Google services



2B LOCs



Example of a program in Python

```
# Solve the quadratic equation ax**2 + bx + c = 0

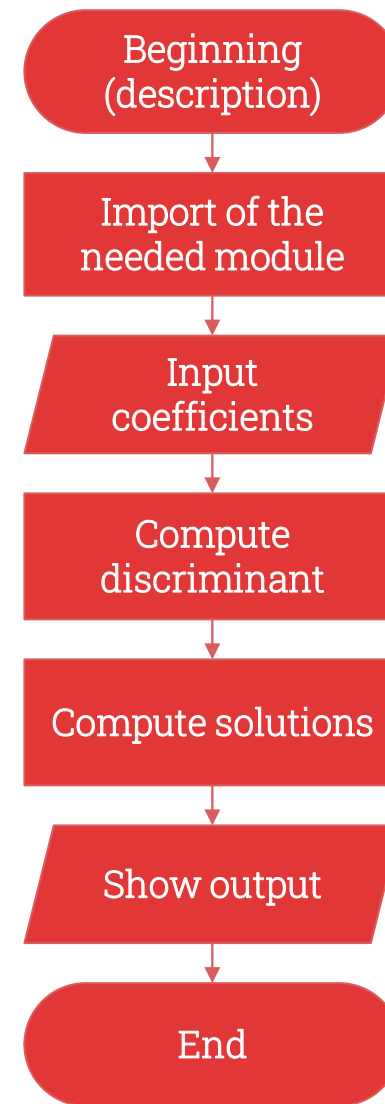
# Import complex math module
import cmath

# Coefficient input from the user
a = float(input('Enter a: '))
b = float(input('Enter b: '))
c = float(input('Enter c: '))

# Compute the discriminant
d = (b**2) - (4*a*c)

# Find two solutions
sol1 = (-b-cmath.sqrt(d))/(2*a)
sol2 = (-b+cmath.sqrt(d))/(2*a)

# show solutions
print('The solutions are ', sol1, ' and ', sol2)
```



What is Python

- Python is a high-level **interpreted, interactive, object-oriented** programming language
- It was conceived by Guido Van Rossum in 1989 and released in 1991, with the aim to correct the defects present in other languages and to take their strengths. The name Python derives from the passion of Guido for the group of British comedians Monty Python
- Since then, Python has grown and has found increasing popularity, first in Web development and, especially in recent years, in the field of data science
- It is an **open source** language, constantly improved by the developer community, and completed by several [libraries](#)

Popularity of programming languages

Feb 2025	Feb 2024	Change	Programming Language		Ratings	Change
1	1			Python	23.88%	+8.72%
2	3	^		C++	11.37%	+0.84%
3	4	^		Java	10.66%	+1.79%
4	2	v		C	9.84%	-1.14%
5	5			C#	4.12%	-3.41%
6	6			JavaScript	3.78%	+0.61%
7	7			SQL	2.87%	+1.04%
8	8			Go	2.26%	+0.53%
9	12	^		Delphi/Object Pascal	2.18%	+0.78%
10	9	v		Visual Basic	2.04%	+0.52%
11	11			Fortran	1.75%	+0.35%
12	15	^		Scratch	1.54%	+0.36%



What is Python used for?

- Python can easily be used for small, large, online and offline projects. Today, the best options to use Python are **web development**, **data analysis** and **simple scripting** (e.g. automating emails or browser actions)
- Python is also used in **machine learning**, **computer vision** (e.g. face detection), **robotics**, **game development**, **web scraping**, and **GUI development**
- Python can be used for creating **prototypes** of applications, that later will be fully developed with more complex or specific languages

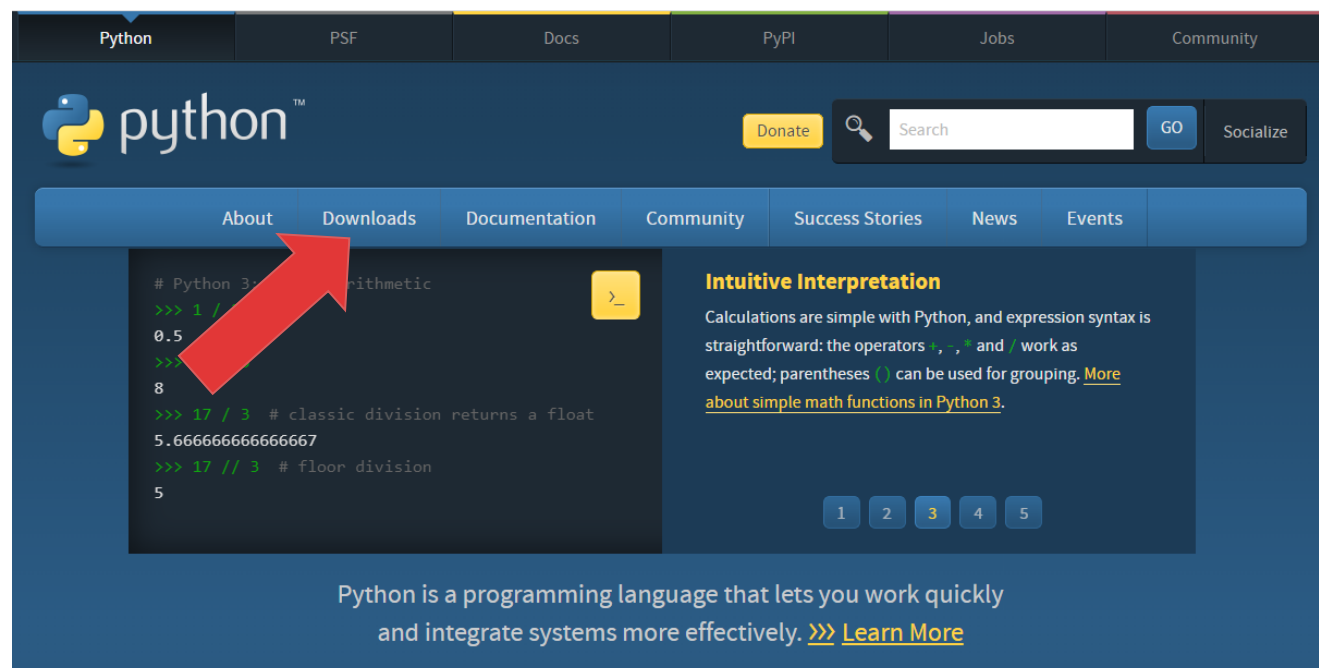


Python success stories

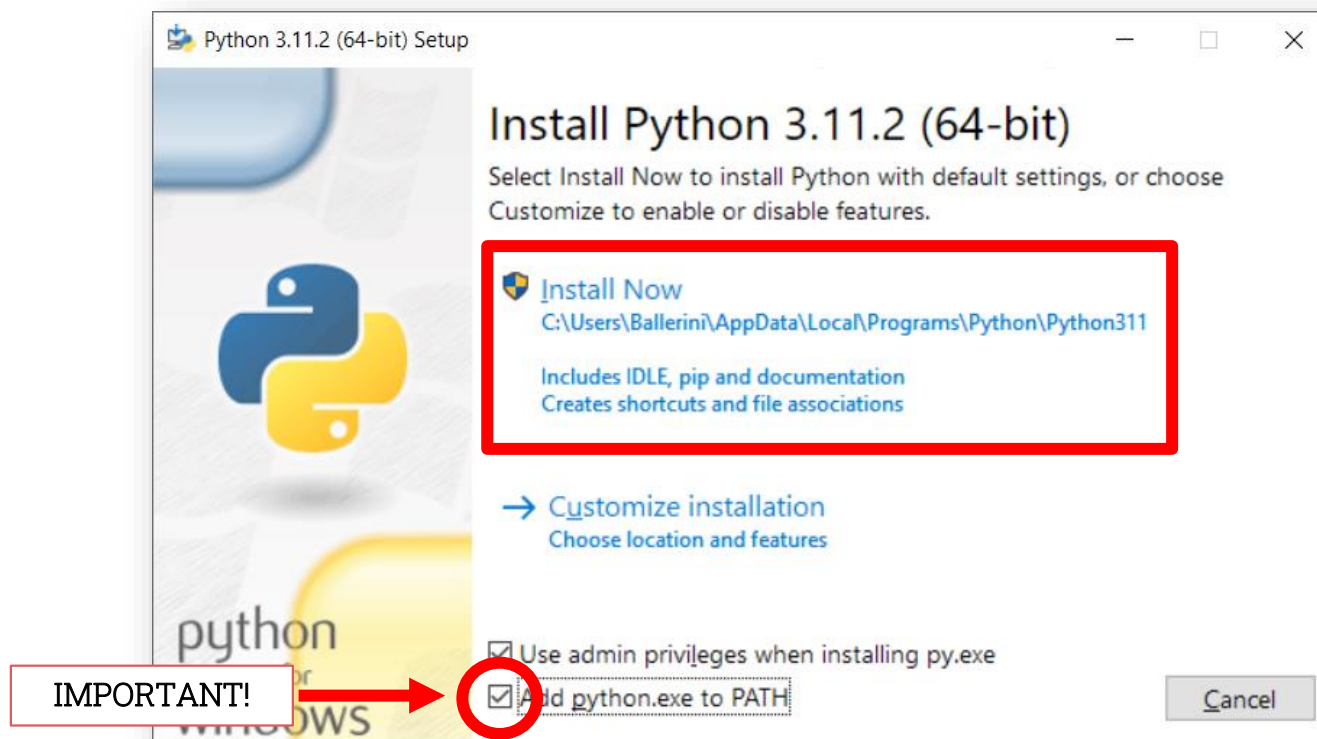
- Contemporary start-ups and big corporations such as **Dropbox** and **Instagram** are using Python to build their sites and services
- **Google** uses Python in its well-known search engine, in several web services and apps
- **YouTube** uses Python to offer several features in its web site
- **Mozilla corporation** (Firefox web browser) uses Python for its web services
- Data management and analysis at **CERN** for experiments to find the first direct evidence of the existence of the Higgs Boson was made with Python
- **ESA** (European Space Agency) is heavily using Python in preparing missions to Mars
- **ILM** (Industrial Light & Magic) has been using Python since mid-1990s to control and automate the production of computer-generated film sequences (e.g. in the second Star Wars trilogy)

Where do we start from?

- The official reference site for download and documentation regarding Python is www.python.org
- The latest version of Python is available in the **Download** section



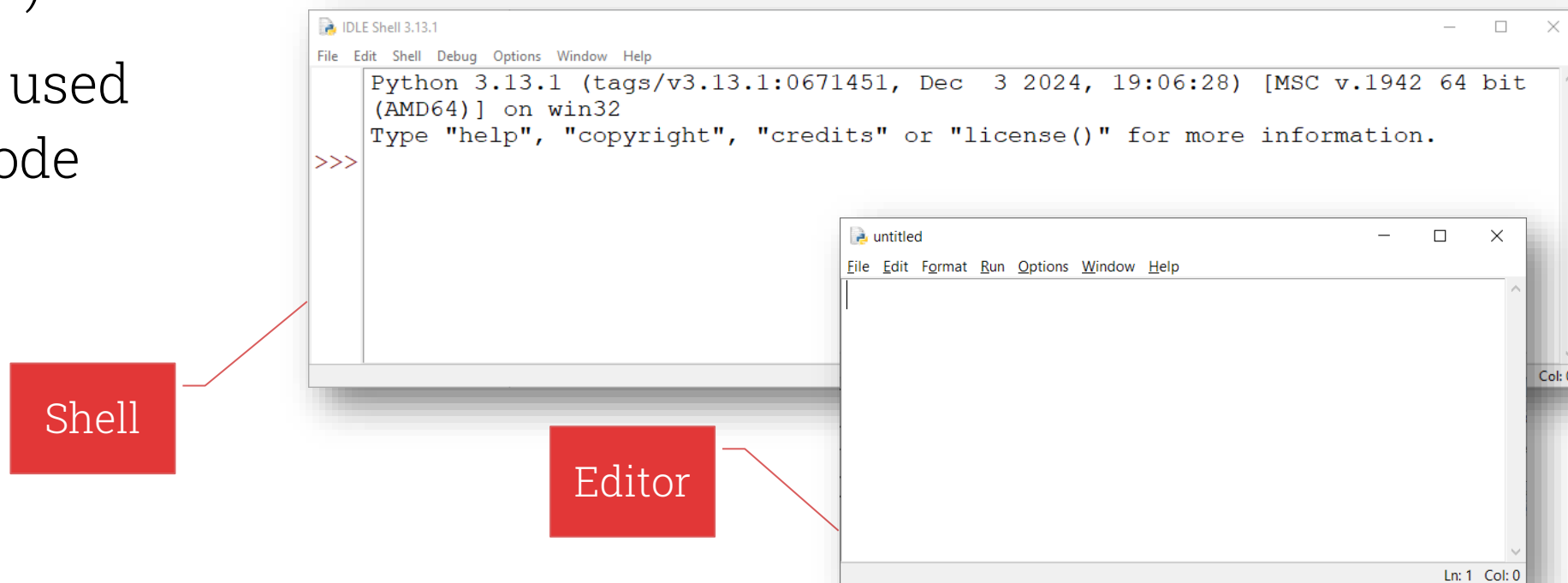
Installing Python



IDLE: editor and shell

IDLE (Integrated Development and Learning Environment) is the Python programming environment, which includes two different components:

- **shell** (or **interpreter**, or **console**), which is used in the interactive mode (or "command line")
- **editor** which is used in the script mode



Opening IDLE

Opening IDLE from the operating system menu will open the **shell** window



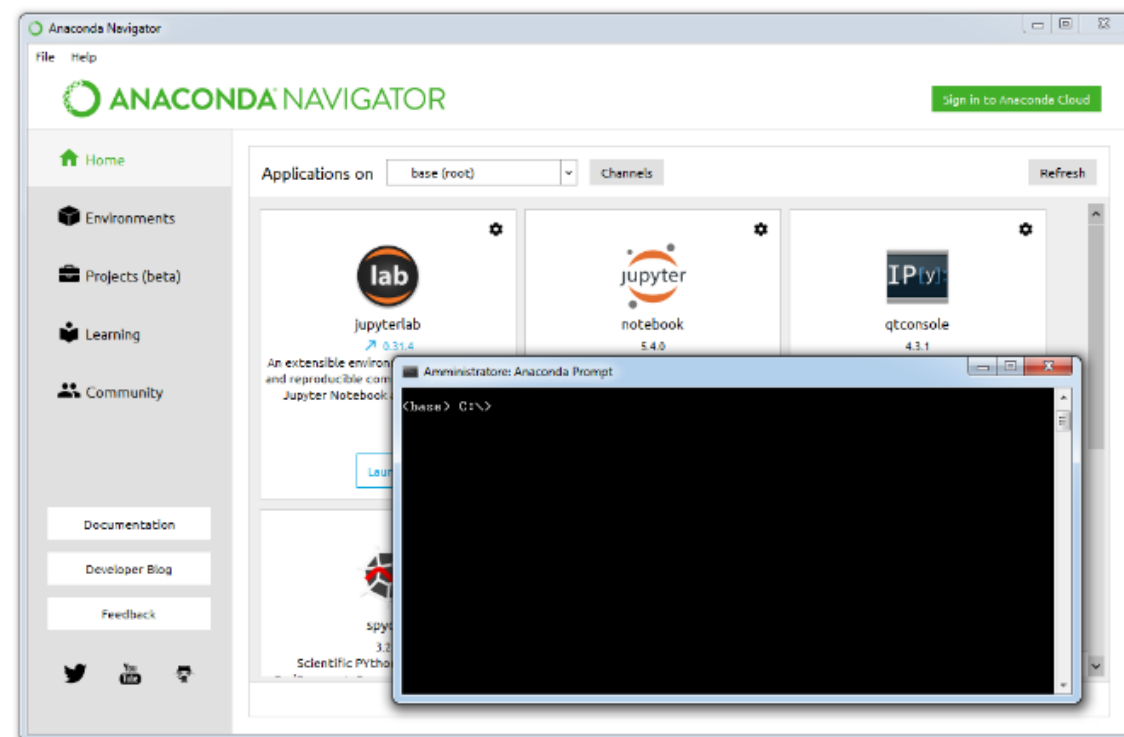
For our first simple examples we will work in the **shell**, afterwards we will move to the editor, that is where we really write programs

Not just IDLE

IDLE is the simplest and most basic tool for using Python, and we will use this in our course

You can use other tools, in addition to IDLE, to develop in Python. The most popular are:

- **Anaconda:**
www.anaconda.com
- **Pythonanywhere:**
www.pythonanywhere.com
- **Atom :**
<https://atom.io/>



Mathematical operators

Symbol	Operation	Description
+	Addition	Adds two numbers
-	Subtraction	Subtracts two numbers
*	Multiplication	Multiplies two numbers
/	Division	Divides two numbers and returns a floating-point number
//	Floor division (aka integer division)	Divides two numbers and rounds down to an integer (*)
%	Modulus	Divides two integers and returns the remainder of the division
**	Exponentiation	Elevates a number to a power

(*) when the result is *negative* it is rounded down to the nearest whole number.



Mathematical expressions

In mathematical expressions, the operations enclosed in parentheses are executed first and then the operator with the highest precedence is applied first:

- Exponentiation
- Multiplication and division
- Addition and subtraction

➡ **PEMDAS** (Parentheses, Exponentiation, Multiplication, Division, Addition, Subtraction)

```
>>> (6 - 3)* (2 + 7) / 3
```

```
9.0
```

```
>>> 6 - 3 * 2 + 7 / 3
```

```
2.3333333333333335
```

Note: numbers in Python are **always** expressed with the **Anglo-Saxon** notation

Order of operations: examples

```
>>> 5 * (3 - 1)
```

10

```
>>> 2**3**2
```

512

```
>>> (22 % 3 + 1)**2
```

4

```
>>> 3*2**3
```

24

```
>>> (2**3)**2
```

64

```
>>> 42//3**2
```

4

```
>>> 42%3**2
```

6

```
>>> (42//3**2 + 42%3**2)**2
```

100



Strings and use of quotation marks

A string in Python can be enclosed in:

- Single quotes: `'goofy'`
- Double quotes: `"goofy"`

In special cases you can also use triple single quotes or triple double quotes

They can be used interchangeably, except where the string contains quotes or quotation marks, so the other character must be used to delimit it

Operations on strings

Normally it is not possible to perform operations on strings, even if their content resembles a number.

There are two exceptions:

- the + operator: performs the concatenation
- the * operator: performs the repetition

Example:

```
>>> 'Python' * 4
```

```
PythonPythonPythonPython
```

Book references

Understanding Python:

Chapters 1, 2

